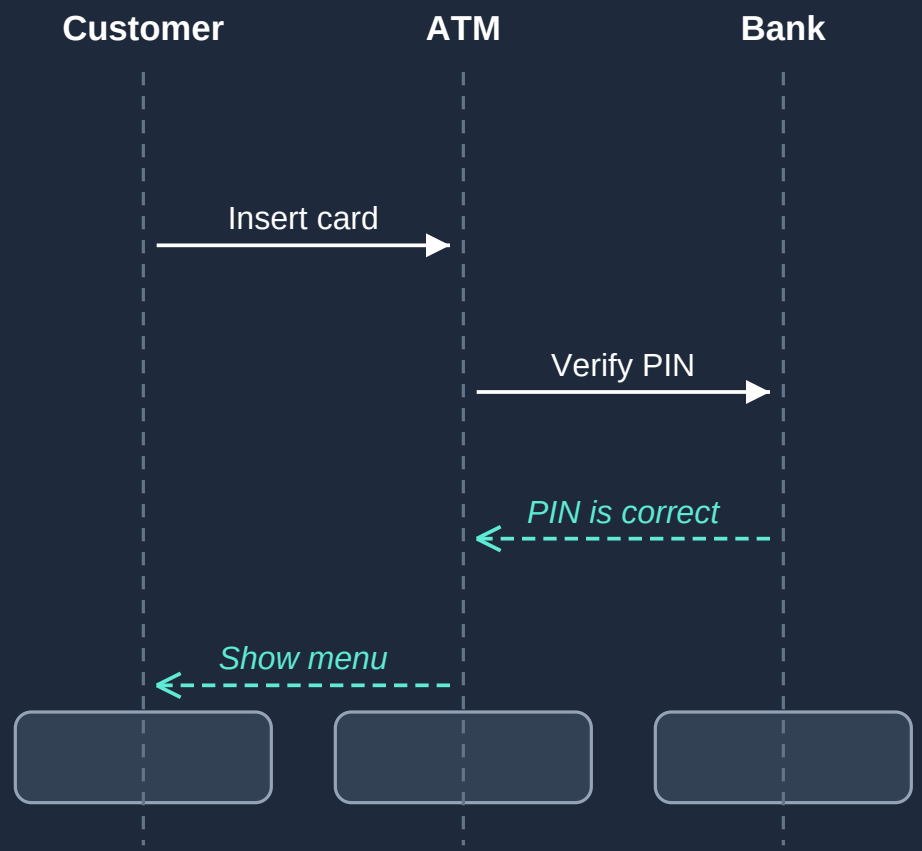
UML | SOFTWARE ENGINEERING

Sequence Diagram

How the parts of a system talk to each other, step by step.



What is a Sequence Diagram?

- A UML diagram that shows how the parts of a system talk to each other over time.
- It answers two questions: who talks to whom, and in what order?
- Time moves from top to bottom.
- One use case can have one or more sequence diagrams.

Use Case Diagram

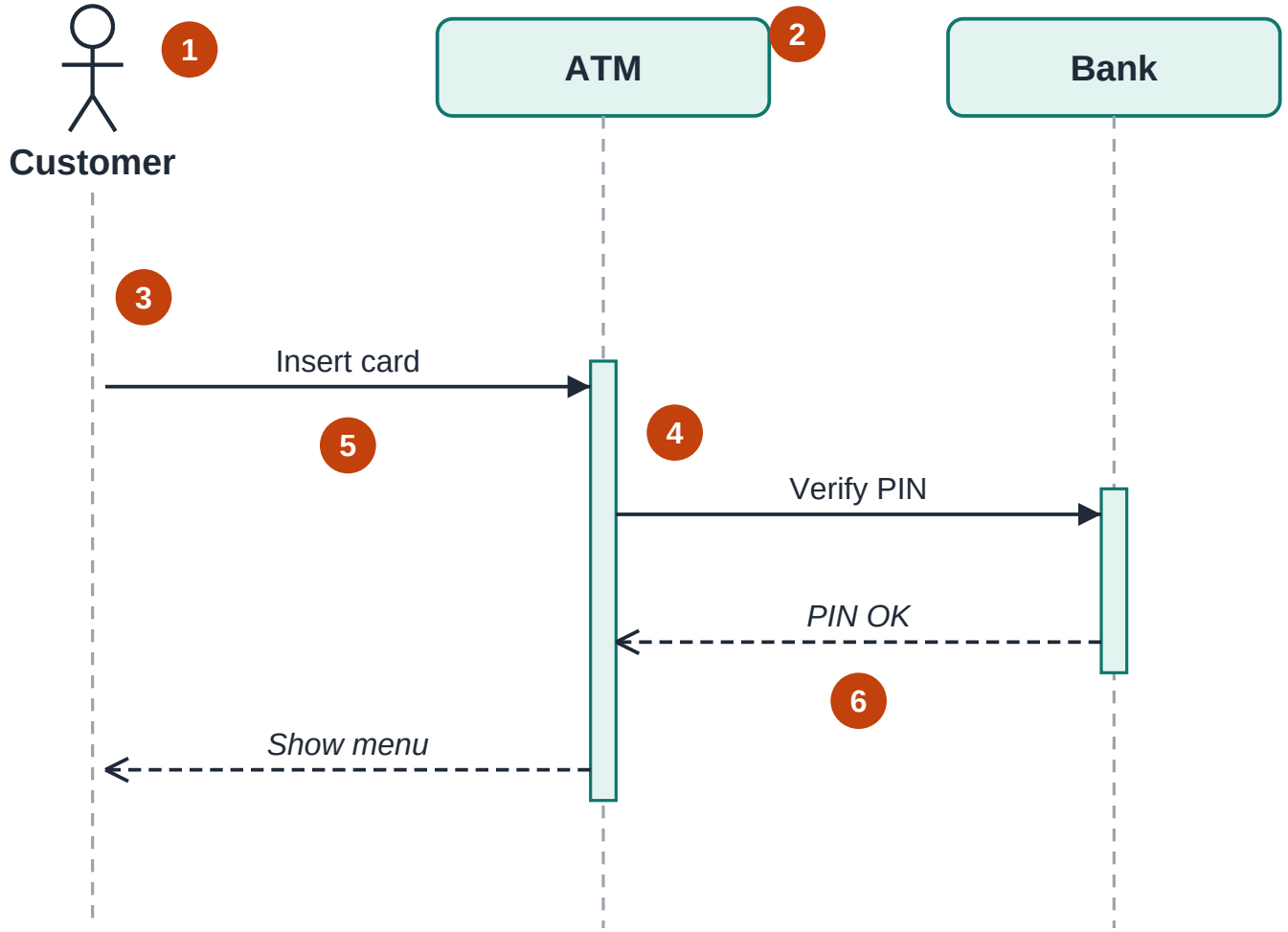
WHAT the system does for its users.

Sequence Diagram

HOW it does it, step by step.

Think of it as a conversation script between the actor and the system.

The parts of a Sequence Diagram



1 Actor
A person or outside system that starts the sequence.

2 Object
A part of the system (ATM, Bank), drawn as a box.

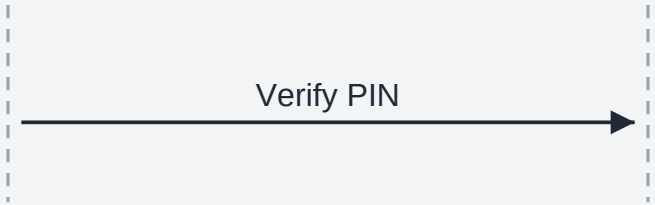
3 Lifeline
The dashed line. It shows time going down.

4 Activation bar
The thin box. It shows when the object is busy.

5 Message
A solid arrow. One object asks another to do something.

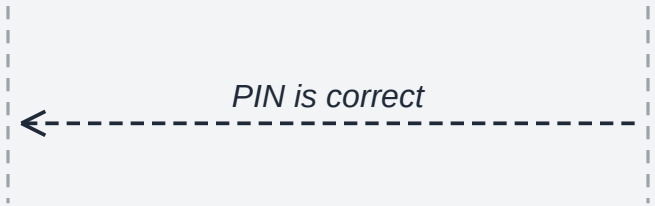
6 Return message
A dashed arrow. The answer coming back.

Types of messages



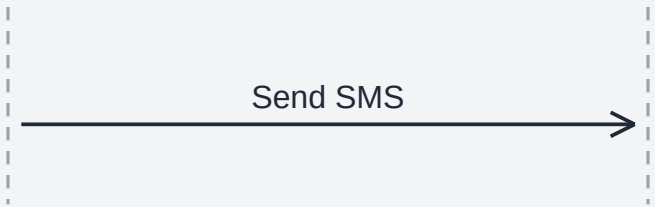
Synchronous message

Solid line, filled arrowhead. The sender waits for the answer.



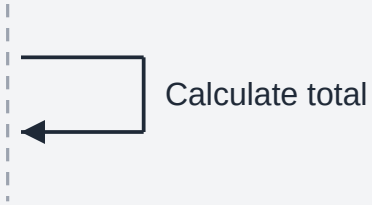
Return message

Dashed line, open arrowhead. The answer going back to the sender.



Asynchronous message

Solid line, open arrowhead. The sender does not wait for an answer.



Self message

An arrow that goes back to the same object. The object calls its own function.

Rules to remember

1 Time goes down

The first message is at the top. The last message is at the bottom.

2 Left to right

Put the actor on the left. Add the other objects in the order they are used.

3 Arrows have direction

Each arrow starts at the sender and points to the receiver.

4 Name messages with verbs

Use actions: Insert card, Verify PIN, Check balance.

5 Answer every request

A request that needs an answer gets a dashed return message.

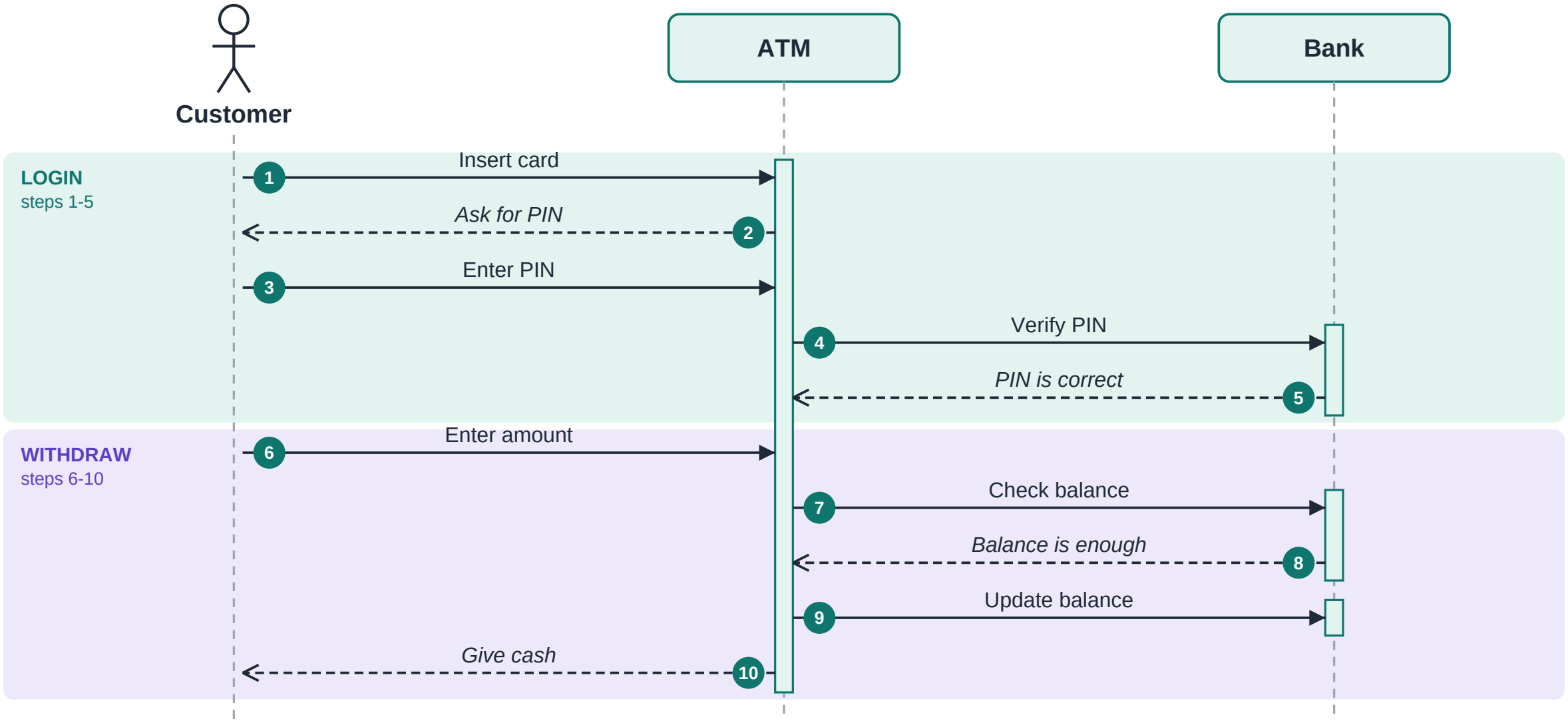
6 One scenario per diagram

Draw one story at a time, for example only Withdraw cash.

6 steps to draw one

1. **Pick one use case.**
   Example: Withdraw cash.
2. **Find the participants.**
   The actor and the objects: Customer, ATM, Bank.
3. **Draw the boxes at the top.**
   Put a dashed lifeline under each box.
4. **Write the story in order.**
   List the steps of the scenario, one by one.
5. **Draw each step as an arrow.**
   From the sender to the receiver, from top to bottom.
6. **Add the answers and check.**
   Every request has a return message, and the order is correct.

Example: Withdraw cash



Walk through the example

Part 1: Login

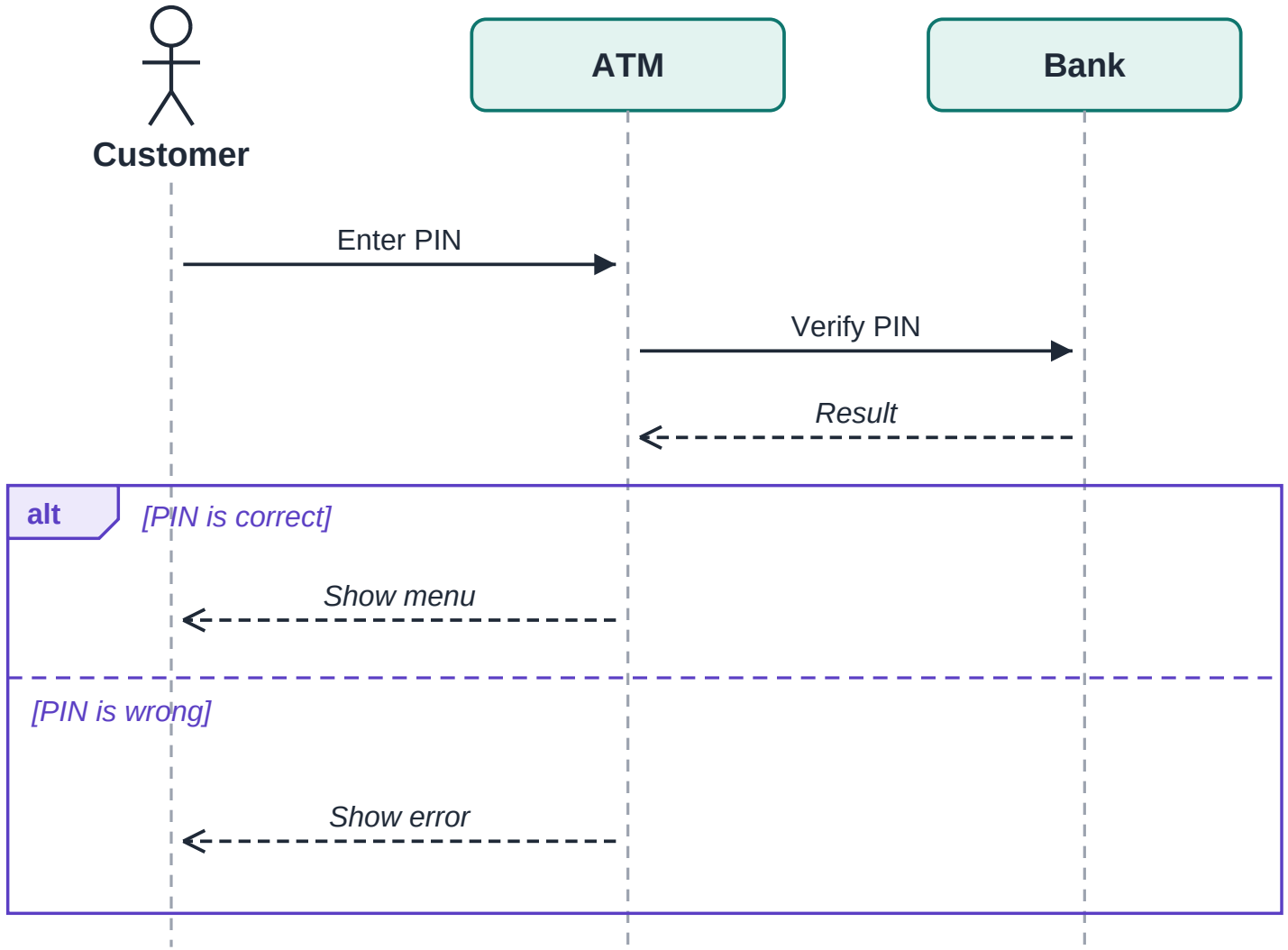
1. The Customer inserts the card into the ATM.
2. The ATM asks for the PIN.
3. The Customer enters the PIN.
4. The ATM asks the Bank to verify the PIN.
5. The Bank answers: the PIN is correct.

Part 2: Withdraw

6. The Customer enters the amount.
7. The ATM asks the Bank to check the balance.
8. The Bank answers: the balance is enough.
9. The ATM asks the Bank to update the balance.
10. The ATM gives cash to the Customer.

Link to the Use Case diagram: Login, Check balance, and Update balance are the same use cases we drew before. Each use case becomes a set of messages.

Fragments: alt, opt, loop



alt = if / else

Only one part runs. Example: PIN correct or PIN wrong.

opt = optional

Runs only if the condition is true. Example: Print receipt [Customer asks].

loop = repeat

Runs again and again while the condition is true. Example: Try PIN [up to 3 times].

opt is the sequence version of Extend.

Common mistakes

Common mistake	How to fix it
✗ **Arrows go up, so the order is wrong.**	✓ Time always goes down. Redraw the arrows in order.
✗ **No return messages.**	✓ Add a dashed arrow when a request needs an answer.
✗ **A solid line for the return message.**	✓ Return messages are dashed, with an open arrowhead.
✗ **Nouns as messages: PIN, Balance.**	✓ Use verbs: Enter PIN, Check balance.
✗ **Too many scenarios in one diagram.**	✓ One scenario per diagram. Draw Withdraw cash and Check balance separately.

Cheat sheet: the symbols

Actor

Starts the sequence.

Object

A part of the system.

Lifeline

Time going down.

Activation bar

The object is busy.

Synchronous

Request. Sender waits.

Return

The answer.

Asynchronous

Request. No waiting.

Self message

Object calls itself.

Fragment

alt, opt, loop.

Your turn: Borrow a book

The scenario

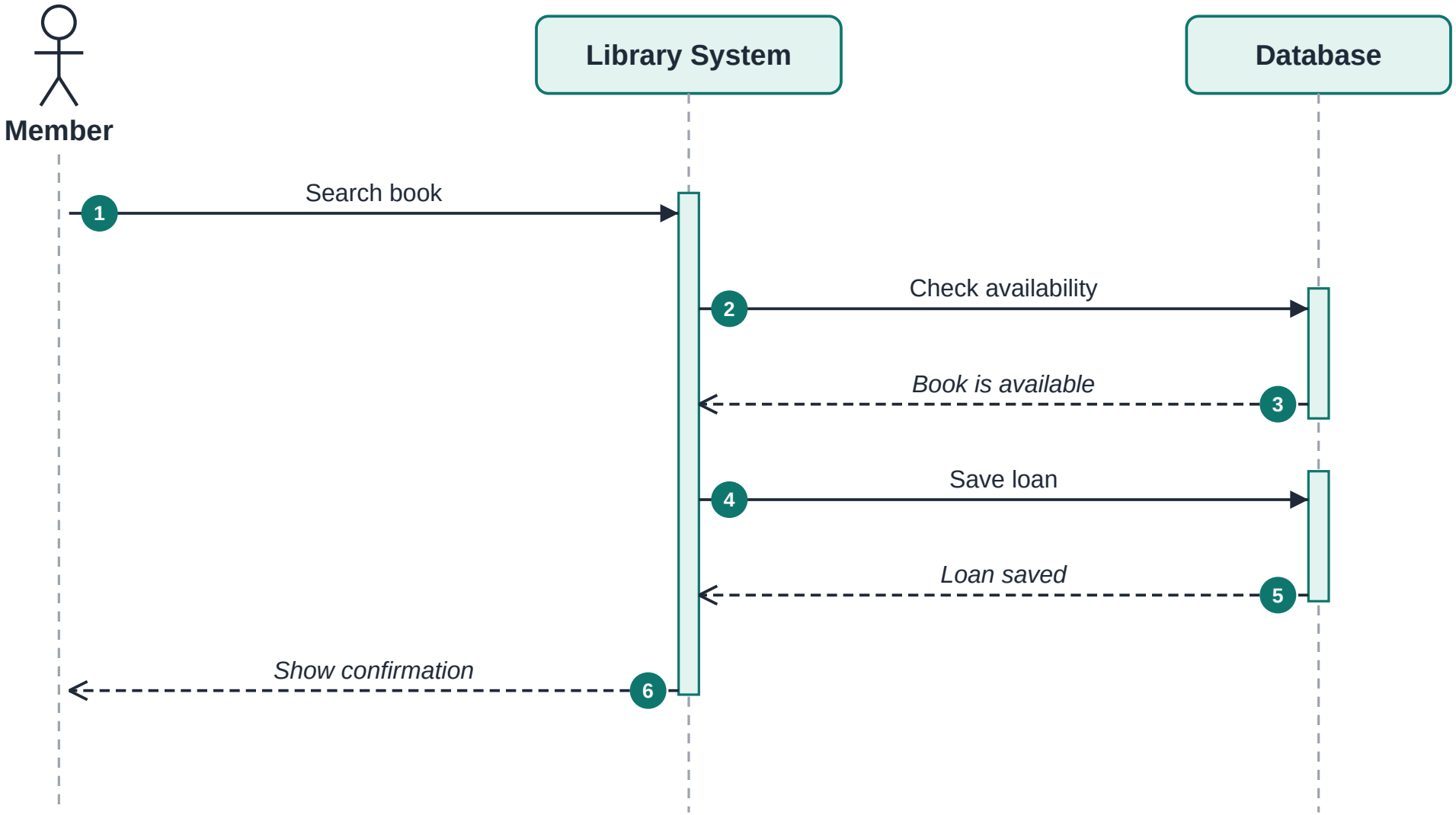
A Member searches for a book. The Library System asks the Database if the book is available. If it is available, the system saves the loan. Then it shows a confirmation to the Member.

Draw the sequence diagram

- Participants: Member, Library System, Database.
- Solid arrows for requests.
- Dashed arrows for answers.
- Time: 5 minutes.

1. **Start with the actor on the left.**
2. **Number your messages.**
3. **Every request needs an answer.**

Answer: Borrow a book



Summary

- ✓ A sequence diagram shows who talks to whom, and in what order.
- ✓ Time goes from top to bottom.
- ✓ Solid arrow = request. Dashed arrow = answer.
- ✓ Fragments (alt, opt, loop) show choices and repeats.
- ✓ Each use case can become a sequence diagram.

Questions?